

NAVAL COMMS, SAFETY, AND SYSTEMS MASTER STUDY GUIDE

Made by: <https://www.answrit.net/>

SECTION 1: ELECTRICAL SAFETY, SHOCK, & TAG-OUT

Shipboard survivability is built upon the foundation of electrical safety and the rigorous execution of tag-out procedures. In the high-density electromagnetic environment of a naval vessel, strict adherence to these protocols is not merely administrative; it is a critical safeguard against personnel casualties and the catastrophic degradation of mission-essential equipment. Failure to isolate energy sources leads to lethal consequences. By establishing clear boundaries and following strict procedures, technicians ensure that maintenance actions do not compromise the integrity of the ship or the lives of the crew.

Element	Definition	Relationship to Safety
Current	The flow of electricity through a path.	The Cause: It is current, not voltage, that causes the sensation and muscular spasms of shock.
Voltage	The force pushing electricity through a circuit.	The Force: Voltage seeks the path of least resistance to ground.
Resistance	The degree to which a body part resists the flow.	The Variable: Damage depends on resistance and the path taken through the body.

Symptoms of Electrical Shock

- **Pale or Bluish Skin:** Signifies severe lack of oxygen or circulation.
- **Unconsciousness:** Immediate loss of awareness following contact.
- **Breathing Difficulties:** Respiratory distress or total cessation.
- **Body Rigidity:** Muscles become stiff, locked, or exhibit "rigidity" in appearance.
- **Loss of Bladder Control:** A common physiological response to high-intensity shock.

Safety Protocols & Personal Protective Equipment (PPE) Hazard mitigation relies on **Avoidance** (jewelry removal, watcher posting) and **Insulation** (non-conductive barriers).

- **Rubber Boots:** Standard issue; insulated up to 600 volts with no exposed metal.
- **Rubber Gloves:** Categorized into Classes 1–4 based on wall thickness and maximum safe voltage.

- **Hearing Protection:** Mandatory exposure to sounds **85 decibels** or higher (plugs or muffs).
- **Eye Protection:** Per OPNAV 5300.23; required for PMs on comms equipment or particulate discharge.
- **Respiratory Safety:** Required in confined spaces where oxygen levels are below **19.5%** or above **21%**.
- **Fall Protection:** Required higher than 5 feet; utilizes self-retracting Y-type lanyards.

RADHAZ & Radiation Safety Radiation Hazards (RADHAZ) encompass dangers to Personnel (**HERP**), Ordnance (**HERO**), and Fueling (**HERF**). Based on [SOURCE_IMAGE_1], specific signage dictates safety actions:

- **Type 1 (Personnel Hazard):** RF hazard exists; keep moving to limit exposure.
- **Type 2 (Command Check):** Beyond this point, check with authority before proceeding.
- **Type 3 (RF Burn):** Warning of burn hazards existing on metallic objects in the area.
- **Type 4 (HERF):** Hazard to fueling operations; check with command.
- **Type 5 (General Warning):** General RF radiation warning.
- **Type 6 (HERO):** Hazard to Ordnance; currents may fire or damage ordnance. Beyond this point, check with authority.

Tag-Out Program Management Managed by the Authorizing Official (CSOOW or EOOW), tag-outs prevent accidental operation of equipment.

Tag Type	Color	Purpose	Required Fields
CAUTION	Yellow	Temporary special instructions/precautionary measures.	6 Fields: Tag ID, Amplifying Instructions, Hanger Sign, Verifier Sign, AO Sign, Contractor.
DANGER	Red	Prohibits operation to protect personnel/hardware.	5 Fields: Tag ID, Hanger Sign, Verifier Sign, AO Sign, Contractor.

Aloft & Fall Safety Aloft work is authorized only by the CO. Work must stop immediately if the ship reaches a **10-degree roll** or if winds exceed **30 knots**. A safety observer is mandatory for any work higher than 5 feet.

Maintaining physical safety through these protocols ensures the technician can focus on the complexities of the electromagnetic environment described in the following section.

SECTION 2: RF PRINCIPLES, PROPAGATION, & ATMOSPHERE

Understanding Radio Frequency (RF) propagation is a strategic requirement for maintaining a ship's radio horizon and ensuring long-distance connectivity. Mastering atmospheric variables allows the Fleet to maintain command and control across vast maritime distances.

Wave Characteristics

- **Velocity:** Speed of travel, equal to the speed of light (186,000 miles per second).
- **Cycle:** One complete repeating pattern of a wave (sine curve up and down).
- **Wavelength:** Physical distance from a point on one cycle to the same point on the next.
- **Frequency:** Cycles completed in one second, expressed in Hertz (Hz).
- **Inverse Relationship: High Frequency = Short Wavelength.**

Atmospheric Stratification The atmosphere consists of layers that dictate signal movement and distance.

Layer	Altitude (Approx.)	Propagation Impact
Troposphere	Surface to 6–10 miles	Lowest layer; most weather occurs here, causing interference.
Stratosphere	10 to 30 miles	Stable wave propagation due to layer stability.
Ionosphere	30 to 250 miles	Contains sub-layers D, E, and F . Solar radiation creates ions that refract or absorb waves (Day = High Ionization; Night = Low).

Propagation Behaviors

- **Reflection:** Waves bounce off flat surfaces like the earth, metallic objects, or the ionosphere.
- **Refraction:** Waves bend as they move between mediums of different density, temperature, or humidity.
- **Diffraction:** Waves bend/spread around solid, dense objects or edges.
- **"So What?":** These behaviors determine if a signal reaches its destination or is lost to the environment.

Wave Components & Attenuation

- **Surface Waves:** Follow the curve of the earth; strength depends on surface **Conductivity** (Water = High/Better range; Rock = Low).
- **Space Waves:** Direct antenna-to-antenna path (Line of Sight); affected by weather.
- **Sky Waves:** Transmitted into the atmosphere and refracted back to earth by the ionosphere for long-range.
- **Attenuation:** Energy loss causing signal strength to decrease; energy is absorbed by the earth.

Frequency Planning: The Window of Usability Technicians use three metrics to identify the usable frequency range:

1. **LUF (Lowest Usable Frequency):** The bottom of the usable window.
2. **MUF (Maximum Usable Frequency):** The top of the usable window.
3. **FOT (Frequency of Optimum Transmission):** The "Optimum" target, providing **90% reliability**.

Once the propagation path is understood, these signals must be digitally manipulated for secure transit.

SECTION 3: DIGITAL COMMS, TIMING, & SIGNAL CONVERSIONS

Digital signal processing is superior to analog for naval operations because it enables high-speed data transfer, advanced error detection, and robust encryption—essentials for modern combat.

Data Transmission Modes

- **Synchronous:** Data transmitted at a rate determined by an associated clock.
- **Asynchronous:** Transmitted without a shared clock; uses start and stop bits (lower speed).
- **Isochronous:** Transmitted without a clock and without start/stop bits; requires high timing consistency.
- **"So What?":** This determines how devices synchronize and if real-time delivery is possible.

The Modulation Matrix Modulation changes carrier properties to carry intelligence/data.

Modulation Type	Category	Properties Changed
AM (Amplitude)	Analog	Changes the height/strength of the signal.
FM (Frequency)	Analog	Changes the number of cycles per second.
PM (Phase)	Analog	Changes the angle/timing of the waveform.
FSK (Freq. Shift Keying)	Digital	Shifts between discrete frequencies (0s and 1s).
PSK (Phase Shift Keying)	Digital	Shifts between discrete phase positions.
QAM (Quadrature Amp.)	Digital	Combines both amplitude and phase shifts.

Signal Classification & Receiver Process

- **RF (Radio Frequency):** On-air signal.

- **IF (Intermediate Frequency):** Internal processing frequency.
- **AF (Audio Frequency):** Audible sound.
- **DC (Direct Current):** Steady power/bias levels.
- **Receiver Path:** Reception → Selection → Detection → Reproduction.

Equipment Roles (DTE vs. DCE)

- **Data Terminal Equipment (DTE):** Endpoint device (e.g., a computer/router).
- **Data Circuit-terminating Equipment (DCE):** Connecting device (e.g., a modem).
- **The Rule: DCE is always the Clock Source.**

Link Quality (Eb/N0) The **Energy per Bit to Noise Density Ratio (Eb/N0)** is the measure of digital link quality. A **higher ratio** results in lower (better) bit error rates.

These fundamentals support the shipboard networks that generate and route mission-critical data.

SECTION 4: NAVAL COMMUNICATIONS FUNCTIONS & SYSTEMS

Naval communications are categorized as **Strategic** (high-level coordination) or **Tactical** (day-to-day fleet operations and movement reports).

Shipboard Network Architecture The Navy utilizes consolidated environments to manage data classification enclaves.

- **CANES:** Consolidated platform that replaces legacy systems like **ISNS** and **SUBLAN**.
- **CENTRIXS:** Coalition information-sharing environment.
- **SCI:** Sensitive Compartmented Information enclave.

ADNS & IP Routing The **Automated Digital Network System (ADNS)** routes shipboard IP traffic toward external transit links. It moves data from the **PT (Plain Text)** side through the **IWE (Inline Network Encryptor)** to the **CT (Cipher Text)** side for secure transmission.

The "Afloat Data Path" Walkthrough (Secret E-mail)

1. **Workstation:** User creates the e-mail.
2. **Switch:** Moves the data to the local server.
3. **Exchange Server:** Recognizes the e-mail as external and queues it.
4. **ADNS (PT side):** Receives plain-text IP data.
5. **IWE/INE:** Encrypts the data (KG-series).
6. **ADNS (CT side):** Routes now-encrypted data toward the RF path.
7. **Crypto/RF Equipment:** Prepares signal for the frequency.
8. **Antenna:** Radiates signal to the satellite.

Shore Architecture & Classification | Command/System | Role/Function | | :--- | :--- | |
NCTAMS | Major regional hub (LANT = Norfolk, PAC = Hawaii). | | **NCTS** | Smaller regional

station providing local infrastructure. | | **STEP** | Tactical Entry Point; connects SATCOM users to terrestrial nets. | | **JFTOC** | Joint Fleet Telecommunications Operations Center; handles **RF/Transit**. | | **NOC** | Network Operations Center; manages **IP Services** (E-mail, DNS, Firewall). |

Enclaves: **NIPRNet** (Unclass), **SIPRNet** (Secret), **JWICS** (TS/SCI), **CENTRIXS** (Coalition).

Network infrastructure relies on the physical hardware units detailed in the next section.

SECTION 5: HARDWARE NOMENCLATURES & ACRONYMS

Nomenclature is vital for maintenance and generating **COMSPOTs** during casualties.

Nomenclature	Description/Function	Study Tip
AN/WSC-3	Standard UHF LOS/SATCOM Transceiver.	Whiskey-3; Fleet workhorse.
AN/USC-61(C) DMR	Digital Modular Radio; software-defined.	Digital Multiple Radios in one.
KY-58 / KYV-5M	Secure voice cryptographic equipment.	K for Key; Voice focus.
KIV-7M	Serial data encryptor.	Standard link encryptor.
KG-175D / KG-250	HAIPE Inline Network Encryptors (IWE).	G for Ground/Network crypto.
PYQ-10 (SKL)	Simple Key Loader; fill device for KEYMAT.	SKL = Simple Key device.
SLM-5650B	Modern SATCOM modem.	Modulates digital for satellite.
SA-2112 / SA-212	Switch Matrices; automatic circuit connection.	SA = Switch Assembly.
AN/SPY-1	Phased-array radar for Aegis.	The "Eyes" of the ship.
AS-2815	Fleet Broadcast Satellite Receive antenna.	Receive-only (SSR-1).
AS-4163	"Stovepipe" dual-channel UHF antenna.	Looks like a chimney pipe.
AS-4172A	"Dog Bowl" mast-clamp JTIDS antenna.	Circular shape like a bowl.

AS-380	"Spider" UHF LOS antenna.	Many "legs" extending out.
R-2368 / R-2557	Standard HF receivers.	R is for Receiver.
TD-1271	UHF DAMA Multiplexer.	DAMA = Demand Assigned.
TD-1603	Fleet Satellite Broadcast Demultiplexer.	Splits one into many.
AN/USC-43(V)	ANDVT (Advanced Narrowband Digital Voice).	Secure voice terminal.
AN/USQ-155	Switch matrix (TS).	Automated signal routing.
SB-4124 / SB-424	Common patch panels (Red/Black).	Physical jack routing.
NAVSSI	Navigation Sensor System Interface.	Sends nav data to systems.
GPNTS	GPS Navigation Timing System.	Source of nav/timing data.
AS-3772	Single whip HF antenna (35ft).	Vertical/Omnidirectional.
AS-1733	Plated-Pairs Antenna.	4 panels; requires clear view.
AS-3818	Gyro-driven tracking SATCOM antenna.	Rotates 360 degrees.
OE-82	Shipboard SATCOM antenna group.	Maintains satellite visibility.
AN/WSC-5(V)	Major Shore Station SATCOM antenna.	Pedestal group; manually oriented.
AS-4543	Mast-clamp "Donut" antenna.	Receive-only HF.
AS-3770B / 3773B	Double whip HF antenna configurations.	Tilted for flight ops.
OA-9242 / 9235	Antenna tilt groups.	Mechanical tilting hardware.
LWCA / OE-367	Wire-rope antenna types.	Long-range wire configs.
MD-1366A/U	EBEM (Enhanced Bandwidth) modem.	High-capacity SATCOM.
MD-1234	Standard SATCOM modem.	Signal formatting.
USQ-144 / K(V)	ADNS variants (Afloat/Shore).	The "Sorter" variants.

NMT	Navy Multiband Terminal.	Protected multiband SATCOM.
CBSP	Commercial Broadband Satellite Program.	Commercial wideband capacity.
NGC2P	Next Generation Command & Control.	CDLMS message support.
AN/UYQ-86	CDLMS (Data Link Management).	Manages TDL/TIC workload.
C2P	Command and Control Processor.	Message routing/time-tagging.
HMI	Human-Machine Interface.	Operator control terminal.
MDU	Multi-Data Unit.	Monitoring/Maintenance/Testing.
ESPG	Embedded System Processor Group.	Heart of the configuration.
C4S A / SYS B	Upper/Lower Chassis designations.	Processing vs. Hardware.
NIMACS	Network Interface Monitor & Control.	V8 console replacing MDU.
TNP	Tactical Network Processor.	Ruggedized PC for STJ functions.
JREAP-C	Joint Range Extension Protocol.	Extending TDL reach.
ELF / VLF / LF	Low Frequency BLOS bands.	Submarine/Long-range.
VHF / UHF / SHF	Standard Comms bands.	Tactical to Wideband.
EHF	Extremely High Freq.	Nuclear survivable/Anti-jam.
SA-212 SAS	Switching Matrix (SAS).	Automated signal routing.
DMR V6	DMR variant with ESPG.	Software-defined architecture.
SSR-1	Receive-only Fleet Broadcast system.	Associated with AS-2815.
STJ / MTJ / UTJ	TADIL-J distribution types.	Sat/Multi/Unicast.
VIXS / SUBLAN	Legacy networks replaced by CANES.	Submarine/Video consolidation.

DISN / NTWAN	Defense/Navy WAN infrastructure.	Global terrestrial backbone.
IWE / INE	Inline Network/Network Encryptor.	Boundary protection.
KGV-136	NMT-only crypto component.	Specific to Multiband Terminal.

Emission Designators (e.g., **37K5 F1D**):

- **37K5**: Bandwidth (37.5 kHz).
- **F (5th)**: Modulation (**F**=Freq, **G**=Phase, **K**=Amp).
- **1 (6th)**: Signal Type.
- **D (7th)**: Data Type.

Hardware units serve as endpoints for the satellite links detailed in the next section.

SECTION 6: SATELLITE ORBITS, BANDS, & TIMELINES

SATCOM is essential for Beyond-Line-of-Sight (BLOS) connectivity in blue-water operations.

Satellite Orbits

- **LEO (400–1,600 miles)**: Low energy; Iridium/Globalstar altitude.
- **MEO (1,500–6,500 miles)**: GPS altitude; synchronous.
- **GEO (22,500 miles)**: Stationary; stays over one spot on Earth.
- **HEO/Molniya (24,000 miles)**: Highly elliptical; high latitude reach.

Frequency Bands (HLL, HLM, HMM) | Band | Name | Characteristics | So What? | | :--- | :--- | :--- | :--- | | **UHF** | Ultra High | **HLL**: High Mobility, Low Capacity, Low Anti-Jam. | Fragile but portable. | | **SHF** | Super High | **HLM**: High Capacity, Low Mobility, Moderate Anti-Jam. | The Fleet "Pipe." | | **EHF** | Extremely High | **HMM**: High Capacity, Moderate Mobility, High Anti-Jam. | Nuclear survivable. |

Operational Timeline & Admin

- **SAR**: Submitted **30–60 days** (min 21) prior to underway.
- **SAA**: Permission granted following a SAR.
- **COMPLAN**: RF specs for ship; done **72 hours** prior.
- **AAR**: Submitted within **10 days** of completion.

Reporting: COMSPOT (30 min window), **CIB** (1=Index, 2=Freq), **2301** (Daily COMMSTAT).

Satellites provide reach, while tactical data links provide the high-speed coordination for combat.

SECTION 7: TACTICAL DATA LINKS, CDLMS, & AEGIS BMD

The Common Tactical Picture (CTP) enabled by real-time data sharing is the backbone of Ballistic Missile Defense (BMD).

Tactical Data Link (TDL) Comparison | Link | Legacy | Path | Purpose | | :--- | :--- | :--- | :--- | |
Link 11 | TADIL-A | HF/UHF | Shared tactical tracks between ships/air. | | **Link 16** | TADIL-J | UHF LOS/SAT | Real-time Joint/Coalition data + Voice. | | **Hawklink** | N/A | UHF/C-Band | Ship-to-Helo (SH-60) exchange. | | **MTJ** | Multicast J | EHF/SHF | One-to-many distribution for BMD. | | **UTJ** | Unicast J | SHF/EHF | Point-to-point J-series delivery. |

CDLMS Architecture (AN/UYQ-86)

- **ESPG:** The **Heart**; handles message processing.
- **HMI:** Operator interface; newer **V8** uses remote session to CND display.
- **Lower Chassis (SYS B):** Signal conversion (**TX:** Digital→Analog; **RX:** Analog→Digital).
- **Upper Chassis (C4S A):** Message routing, time-tagging, and destination selection.

Aegis BMD Process: **Detect** (AN/SPY-1) → **Track** (Maintain contact) → **Intercept** (SM-3).

- **AN/SPY-1:** Phased-array; tracks 100+ targets simultaneously.
- **SM-3:** Midcourse interceptor. **SM-2:** Terminal phase defense.

Watchstations: **TAO** (Lead), **BMDO** (Liaison), **TIC** (Link manager), **RSC** (Radar ops), **MSS** (Missile supervisor), **Link Response Team** (IT, FC, ET, OS, GM) for rapid repair.

End-to-end communications provide the warfighter the right information at the right time to ensure mission success.